Module 1 — Introduction to Artificial Intelligence

1.1 What is Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building machines that can perform tasks requiring human intelligence.

These tasks include:

- reasoning
- learning
- perception
- language understanding
- problem solving
- decision making

AI systems attempt to simulate intelligent behavior by processing data and applying algorithms.

AI combines knowledge from several disciplines including:

- Computer Science
- Mathematics
- Psychology
- Cognitive Science
- Philosophy
- Linguistics

1.2 Goals of Artificial Intelligence

The primary goal of AI is to build machines that can:

1. **Think like humans**
2. **Act like humans**
3. **Think rationally**
4. **Act rationally**

These four perspectives are often used to classify AI systems.

1.3 Applications of AI

AI is widely used across many domains:

Healthcare

- Disease diagnosis
- Medical image analysis
- Drug discovery

Finance

- Fraud detection
- Algorithmic trading
- Risk assessment

Transportation

- Self-driving cars
- Traffic prediction
- Route optimization

E-commerce

- Recommendation systems
- Customer support chatbots

Education

- Intelligent tutoring systems
- Automated grading

1.4 Intelligent Agents

An **agent** is any entity that perceives its environment through **sensors** and acts upon that environment using **actuators**.

Example:

Human Agent

Sensors:

- eyes
- ears

Actuators:

- hands
- legs
- mouth

Robot Agent

Sensors:

- camera
- GPS
- sonar

Actuators:

- motors
- robotic arms

1.5 PEAS Representation

PEAS stands for:

Performance measure
Environment
Actuators
Sensors

Example: Taxi driver agent

Performance Measure:

- safety
- speed
- comfort
- profit

Environment:

- roads
- traffic
- pedestrians

Actuators:

- steering
- brakes
- accelerator

Sensors:

- cameras
- GPS

- speedometer

This framework helps define the **task environment of an AI system**.

1.6 Properties of Task Environments

AI environments are classified using several properties.

Fully Observable vs Partially Observable

Fully Observable:
All information about the environment is available.

Example:
Chess

Partially Observable:
Some information is hidden.

Example:
Poker

Deterministic vs Stochastic

Deterministic:
Outcome of actions is predictable.

Stochastic:
Outcome involves randomness.

Example:
Weather prediction.

Episodic vs Sequential

Episodic:
Each action is independent.

Sequential:
Actions affect future states.

Static vs Dynamic

Static:
Environment does not change while agent thinks.

Dynamic:
Environment changes continuously.

Discrete vs Continuous

Discrete:
Finite states.

Continuous:
Infinite possibilities.

Example:
Robot motion.

1.7 Problem Solving in AI

Problem solving involves finding a sequence of actions that lead from the **initial state to the goal state**.

Components of problem solving:

1. Initial state
2. Actions
3. Transition model
4. Goal state
5. Path cost

1.8 State Space Representation

A **state space** represents all possible states of a problem.

Example:
8-puzzle problem.

Nodes represent states.

Edges represent actions.

Search algorithms explore the state space to reach the goal.

1.9 Search Techniques

Search methods are divided into two categories.

Uninformed Search

These methods do not use domain knowledge.

Examples:

Breadth First Search (BFS)

Depth First Search (DFS)

Uniform Cost Search (UCS)

Informed Search

Uses heuristics to guide the search.

Examples:

Greedy Best First Search

A* Algorithm

Heuristic functions estimate the distance to the goal.

1.10 Game Playing in AI

AI can play games using search algorithms.

Example: Chess.

Minimax algorithm assumes the opponent plays optimally.

The algorithm tries to maximize the minimum possible gain.

Alpha-Beta Pruning

Improves Minimax efficiency.

It removes branches that cannot influence the final decision.

Benefits:

- faster computation
- reduced search space

Module 2 — Knowledge Representation

2.1 What is Knowledge Representation

Knowledge Representation (KR) is the method used to represent information about the world so that machines can reason about it.

A **knowledge base** stores:

facts
rules

AI systems use inference mechanisms to derive new knowledge.

2.2 Knowledge Based Agents

A knowledge based agent works using two operations:

TELL

Adds new information to the knowledge base.

ASK

Queries the knowledge base.

This allows the system to reason and decide actions.

2.3 Levels of Representation

Knowledge can be represented at two levels.

Knowledge level:
Conceptual description of facts.

Symbol level:
Actual implementation in the system.

2.4 Properties of Knowledge Representation Systems

A good KR system should provide:

Representational adequacy
Inferential adequacy
Inferential efficiency
Acquisitional efficiency

These properties ensure the system can represent and manipulate knowledge effectively.

2.5 Approaches to Knowledge Representation

Several approaches exist.

Logical Representation

Uses formal logic to represent knowledge.

Example:

All humans are mortal

Human(Socrates)

Therefore:

Mortal(Socrates)

Semantic Networks

Graph based representation.

Nodes represent objects.

Edges represent relationships.

Example:

Dog → is-a → Animal

Frames

Frames represent structured knowledge using slots.

Example:

Frame: Car

Attributes:

- color
- engine
- manufacturer

Rule Based Systems

Knowledge represented as rules.

IF condition THEN action

Example:

IF temperature > 38°C
THEN fever = true

2.6 Predicate Logic

Predicate logic represents relationships using predicates.

Example:

man(Marcus)

∀x (man(x) → mortal(x))

This means all men are mortal.

2.7 Forward and Backward Reasoning

Forward reasoning:

Start from facts and derive conclusions.

Backward reasoning:

Start from goal and work backwards.

Used in expert systems.

Module 3 — Probabilistic Reasoning and Expert Systems

3.1 Uncertainty in AI

Real world problems involve uncertainty.

Example:

Medical diagnosis

Weather prediction

AI systems use probabilistic methods to reason under uncertainty.

3.2 Bayesian Networks

A Bayesian network is a probabilistic graphical model.

Components:

Nodes → variables
Edges → dependencies

Each node has a conditional probability table.

These networks model probabilistic relationships between variables.

3.3 Markov Blanket

The Markov blanket of a node includes:

- its parents
- its children
- parents of its children

Given the Markov blanket, the node becomes independent of the rest of the network.

3.4 Dempster-Shafer Theory

Dempster-Shafer theory handles uncertain evidence.

Instead of exact probabilities, it uses:

Belief

Plausibility

Belief represents the degree of evidence supporting a hypothesis.

Plausibility represents how possible the hypothesis could be.

This method is useful when information is incomplete.

3.5 AI Planning

Planning is the process of determining actions required to achieve goals.

Example:

Robot navigation.

Planning involves:

Initial state
Goal state
Actions

Types of Planning

Goal Stack Planning

Hierarchical Planning

Conditional Planning

Probabilistic Planning

Temporal Planning

3.6 Expert Systems

Expert systems simulate the decision making of human experts.

Components:

Knowledge Base
Inference Engine
User Interface

Applications include:

medical diagnosis
fault detection
financial decision making

Module 4 — Natural Language Processing

4.1 What is NLP

Natural Language Processing allows computers to understand and generate human language.

Applications include:

chatbots
translation
summarization
question answering

4.2 NLP Components

NLP involves multiple levels of analysis.

Morphological Analysis
Syntactic Analysis
Semantic Analysis
Discourse Integration
Pragmatic Analysis

These steps help machines understand language structure and meaning.

4.3 Text Processing

Raw text must be processed before analysis.

Key steps include:

Tokenization
Normalization
Stop word removal
Stemming
Lemmatization

4.4 Feature Representation

Text must be converted into numerical form.

Common techniques:

Bag of Words
TF-IDF
N-grams
Word vectors

These representations allow machine learning models to process text.

4.5 Transformers

Transformers are deep learning models designed for sequence processing.

Key concept:

Attention mechanism

Attention allows models to focus on relevant parts of a sentence.

Transformers power modern models such as:

BERT
GPT
T5

Module 5 — Generative AI

5.1 What is Generative AI

Generative AI refers to models that create new data similar to training data.

Examples:

text generation
image generation
music generation
code generation

5.2 Large Language Models

LLMs are trained on massive datasets.

Capabilities:

conversation
translation
summarization
question answering

Examples:

GPT models
BERT
LLaMA

5.3 Prompt Engineering

Prompt engineering improves AI responses.

Methods include:

Zero-shot prompting
Few-shot prompting
Chain-of-thought prompting

5.4 Fine Tuning

Fine tuning adapts pretrained models to specific tasks.

Example:

medical chatbot

legal document analysis

5.5 Explainable AI

Explainable AI focuses on making AI decisions understandable.

This improves:

trust
transparency
accountability

Module 6 — AI Applications and Tools

6.1 Agentic AI

Agentic AI uses autonomous agents capable of:

planning
reasoning
acting

Agents can complete complex tasks independently.

6.2 Responsible AI

Responsible AI ensures ethical development of AI systems.

Key aspects:

fairness
transparency
safety
privacy

6.3 AI Development Frameworks

Popular frameworks include:

PyTorch
TensorFlow
HuggingFace

These frameworks provide tools for training AI models.

6.4 AI Prompting Methods

Zero Shot

Model receives no examples.

Few Shot

Model receives small number of examples.

Chain of Thought

Model generates reasoning steps.

6.5 Retrieval Augmented Generation (RAG)

RAG combines:

information retrieval
language generation

Steps:

1 Retrieve relevant documents
2 Provide documents to LLM
3 Generate answer using retrieved knowledge

Benefits:

reduces hallucinations
improves factual accuracy